

## Problem

Consider a graph  $G$  with edges  $E$  and vertices  $V$ , and subgraphs  $G_1, \dots, G_k \subset G$ . Let  $G_i = (E_i, V_i)$  for these subgraphs. For now let us suppose the graph is unweighted and undirected. The subgraphs need not cover  $G$  and can overlap. Each player  $i$  knows  $G$  and  $G_i$ , but not the other player's graphs. (This is akin to a map of a city, where each player is a different regional taxi company.) The goal is, given some starting vertex  $v \in V$ , and some end vertex,  $w \in V$  to:

1. Identify if it is possible to construct a path  $v, v_1, \dots, v_l, w$  such that every edge on the path is in  $\bigcup_{i=0}^k E_i$ .
2. If such a path exists, find the minimum number of edges needed to get from  $v$  to  $w$ , while using only edges belonging to  $\bigcup_{i=0}^k E_i$ .

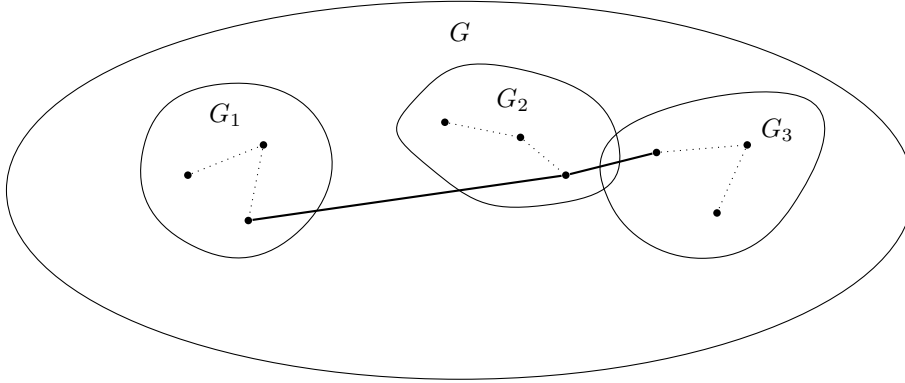


Figure 1: A valid path from  $v$  to  $w$ .

## Lower bound sketches and ideas

For item 1, we think this would require computing  $\binom{k}{2}$  pairwise set disjointness in the worst case, to determine where the intersections of each  $G_i$  are, at which point each player could on their own determine the possible paths between each of the subgraphs that intersect with their own.

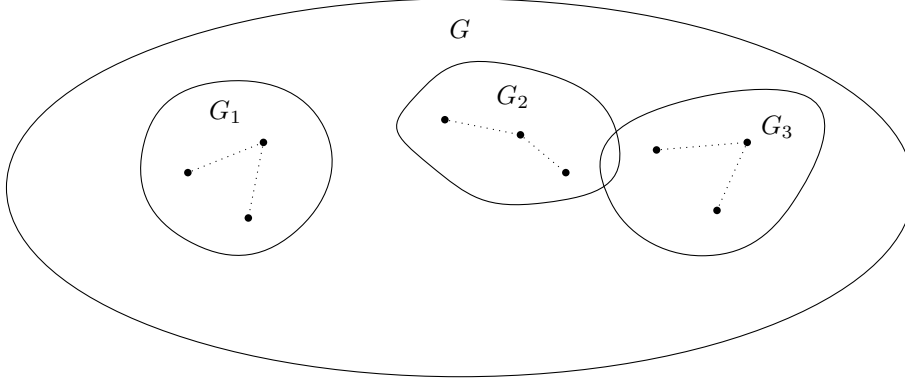


Figure 2: Worst case for item 1: Disjoint Subsets

For item 2, we would be required to compute  $\binom{k}{2}$  pairwise intersections, since in order to minimize the path we would need to know all possible paths that could connect  $v$  and  $w$ . To find the shortest path, each player works backwards and sends the distances through their internal vertices to the players they intersect with, and each player will attempt to identify if a longer path can be dropped whenever they receive this list.